

REVIEW

Personalized Nutrition for Hypertension: Current Evidence and Future Directions

Phoebe Cheong¹ ID, MSc*; Cass Mangan² ID, MSc*; Erin D. Clarke³ ID, PhD; Jordan Stanford⁴ ID, PhD; Murray J. Cairns⁵ ID, PhD; Clare E. Collins⁶ ID, PhD; Francine Z. Marques⁷ ID, PhD†; Matthew Snelson⁸ ID, PhD†

ABSTRACT: Hypertension is a leading risk factor for cardiovascular disease, with lifestyle interventions, particularly diet, playing a pivotal role in its management. Recent advancements in personalized nutrition that tailor dietary recommendations based on an individual's unique biology (eg, genetics, microbiome) and lifestyle factors can potentially improve blood pressure control. Indeed, emerging evidence suggests that personalized nutrition can result in more effective and sustained blood pressure reductions than general dietary guidelines. This review explores the current evidence supporting personalized nutrition as a novel strategy for managing hypertension, while leveraging established approaches for controlling glycemia. We evaluate studies integrating genetic markers and genomics, gut microbiome profiles, and metabolic phenotypes to design personalized dietary interventions to reduce blood pressure. Specifically, we discuss how nutritional factors, including sodium and fiber intake and diets such as the Dietary Approaches to Stop Hypertension, interact with an individual's genetic predisposition and gut microbiota composition to influence blood pressure outcomes. However, challenges remain in translation to clinical practice, including the need for larger, long-term trials in real-world settings, accessibility of advanced dietary assessments, considerations of patient adherence, and equity. We highlight the potential of personalized nutrition to refine hypertension treatment and call for further research to optimize its implementation at the public health and clinical practice levels. Particularly in low- and middle-income countries, where hypertension prevalence is higher, cost is an important consideration for successful implementation. However, some personalized treatments focusing on diet are cost-effective and should be prioritized. Ultimately, a personalized nutrition approach may offer a powerful tool for mitigating the global burden of hypertension through targeted, individualized dietary strategies.

Key Words: blood pressure ■ diet ■ genetics ■ hypertension ■ metabolites ■ microbiome ■ sodium

Hypertension is a highly preventable cause of death, with the World Health Organization estimating that 76 million deaths could be averted between 2023 and 2050 with optimal blood pressure (BP) management.¹ High BP affected 1.3 billion people in 2019, with less than half receiving a diagnosis and only 21% having it adequately controlled.² Hypertension predisposes individuals to a myriad of deadly conditions, including, but not limited to, coronary disease, heart failure, stroke, peripheral artery disease, and renal failure.³ In addition, hypertension is often associated with metabolic disorders, such as dyslipidemia, insulin resistance, and inflammation, all of which occur concurrently with obesity.⁴

Although nonmodifiable risk factors such as age, sex, and genetics have a substantial influence on BP, lifestyle modifications are considered first-line therapy before or in conjunction with pharmaceutical interventions.⁵

Dietary interventions are considered the most effective lifestyle modifications for lowering BP, with substantial evidence supporting the antihypertensive effects of the Dietary Approaches to Stop Hypertension (DASH) and the Mediterranean diet.⁵ Both these dietary patterns focus on increasing nutrient-rich foods that are high in fiber, including fruits, vegetables, whole grains, and legumes, with the DASH diet being one of the few dietary patterns to have a specific sodium target. Improving diet

Correspondence to: Francine Marques, PhD, Hypertension Research Laboratory, Victorian Heart Institute, Level 2, Victorian Heart Hospital, Monash University, 631 Blackburn Rd, Clayton, Victoria 3168, Melbourne, Australia, Email francine.marques@monash.edu; Matthew Snelson, PhD, Gut Homeostasis in, Cardiometabolic Health Laboratory, Nutrition & Health Innovation, Research Institute, Edith Cowan University, 270 Joondalup Dr, WA 6027, Joondalup, Australia, Email m.snelson@ecu.edu.au

*P. Cheong and C. Mangan contributed equally as first authors.

†F.Z. Marques and M. Snelson contributed equally as senior authors.

© 2026 American Heart Association, Inc.

Circulation: Genomic and Precision Medicine is available at www.ahajournals.org/journal/circgen

Nonstandard Abbreviations and Acronyms

CutC and CutD	choline trimethylamine lyase enzymes C and D
DASH	Dietary Approaches to Stop Hypertension
DASH-JUMP	DASH–Japan Ube Modified Diet Program
DBP	diastolic blood pressure
ENaC	epithelial sodium channel
GPR	G protein–coupled receptor
HCS	Hunter Community Study
INTERMAP	International Study of Macro/Micronutrients and Blood Pressure
PCSK9	proprotein convertase subtilisin/kexin type 9
PES	Pharmagenic Enrichment Score
PPGR	postprandial glucose response
PREDICT	Predicting Responses to Dietary Composition Trial
PRS	polygenic risk scores
PTGS	prostaglandin-endoperoxide synthase
SBP	systolic blood pressure
SCFA	short-chain fatty acid
SSaSS	Salt Substitute and Stroke Study
UKBB	UK Biobank

quality aligns with the American Heart Association's recent *Food Is Medicine* roadmap.⁶ Although dietitians already personalize nutrition care through Medical Nutrition Therapy, achieving long-term behavior change and patient adherence remains challenging. Medical Nutrition Therapy is an evidence-based, highly individualized approach provided by a registered dietitian that uses nutrition assessment, intervention, and monitoring, incorporating medical history, biochemistry, clinical parameters, preferences, and health goals.⁷ Effective nutrition therapy often requires time, ongoing monitoring, and adjustments. This can create barriers to continued engagement, including financial costs of repeat appointments and the perception that dietary interventions are ineffective when results are not immediate.⁸ There is now an opportunity to strengthen personalized nutrition approaches and address these challenges by harnessing emerging data, such as genetic profiles, gut microbiome composition, and continuous or real-time monitoring of diet and health, to guide more precise and timely dietary recommendations. Although biological data form an important component of personalization, effective personalization must also account for behavioral and contextual factors such as food preferences, cultural context,

meal timing, and eating behaviors. This evolving field is commonly referred to as personalized nutrition when focused on individuals, or precision nutrition when targeting specific population subgroups (Figure 1).⁹ Salt sensitivity tends to be greater among women¹⁰ and Black individuals.¹¹ Among hypertensive women, the increase in systolic BP (SBP) after a salt challenge appears more pronounced in Black women compared with White women.¹² Thus, recommendations around dietary sodium specifically for Black women with hypertension exemplify precision nutrition, as it tailors dietary guidance to a defined subgroup. Tailoring dietary fiber subtypes based on an individual's gut microbiota profile represents personalized nutrition. These approaches contrast with general healthy eating guidelines—such as those promoting fruit and vegetable intake—which would be considered population nutrition.⁹

Personalized nutrition is cost-effective in the context of obesity, diabetes, and impaired glucose tolerance.¹³ For multifactorial diseases such as hypertension, intervention-based precision nutrition for target groups is effective in the context of averted disability-adjusted life years not exceeding the average national gross domestic product per capita, as reviewed elsewhere.¹⁴ However, with less than half of adults with hypertension diagnosed and treated globally,¹ there is an urgent need to deliver effective nonpharmacological treatments to address untreated hypertension. This review examines the potential of personalized nutrition to lower BP through integrated insights from gut microbiota, genomics, and diet, while highlighting innovative strategies and identifying key research gaps and challenges.

DIETARY ASSESSMENT METHODS

For personalized nutrition to be successful, accurate assessments of food intake at the individual level are required. Acquiring dietary intake data is notorious for reporter bias, with underreporting or overreporting common.¹⁵ The 3 primary data collection methods for diet are prospective food records, retrospective food recalls, and food frequency questionnaires (FFQ; Table 1). Food records involve real-time logging of all food and beverages consumed, including type, amount, preparation method, time, brand names, and additional condiments or snacks. By capturing this information immediately after consumption, food records can reduce recall bias and provide valuable insights into eating patterns, meal timing, and portion sizes, enabling precise nutrient analysis. However, they may be subject to the Hawthorne effect,¹⁶ where participants may modify their diet because they are aware of being observed as part of the study, rather than in response to the intervention itself. Weighed food records have been considered the imperfect gold standard and involve weighing all food and beverages

Term	Definition	Examples
Population Nutrition	Broad recommendations that will promote health in the majority of the population.	Iodine fortification of commonly consumed food products to prevent deficiencies. Public health education campaign to increase fruit and vegetable intake (e.g. 5 A Day, Go for 2 & 5).
Precision Nutrition	Recommendations that will promote health in a population subgroup, based on subgroup characteristics.	Fortification requirements for infant formula. Calcium intake recommendations for post menopausal women. Recommendations to limit salt intake for people with hypertension.
Personalised Nutrition	Recommendations that will promote health in an individual based on individual characteristics.	Culturally appropriate low phenylalanine diet for an individual with phenylketonuria. Specific dietary fibre intervention based on gut microbiota composition. Tailored sodium intake for someone who has salt-sensitive hypertension.

Figure 1. The difference between population, precision, and personalized nutrition.

Population nutrition refers to general dietary guidelines designed for broad public health applications. Precision nutrition considers characteristics of a subgroup of the population to develop targeted dietary recommendations for specific groups, which considers factors such as life stage and disease status. Personalized nutrition tailors dietary advice to an individual's unique characteristics, including metabolic responses and personal preferences.

consumed over a set number of days. However, this can be time-consuming and burdensome, resulting in misreporting, incomplete entries, or alterations in usual dietary patterns.¹⁵ A food recall involves individuals describing all foods and beverages consumed over a specific period, typically the past 24 hours, through detailed interviews using multiple passes to facilitate recall and avoid both omissions and intrusions.¹⁵ Although this method can improve the accuracy of dietary data for personalized nutrition plans, its limitations include the need for multiple days to approximate an individual's usual food and nutrient intakes, reliance on memory, potential misreporting, and underreporting of unhealthy foods.¹⁵ An FFOQ is a dietary assessment tool designed to evaluate an individual's usual intake based on reporting intake frequency from a list of commonly consumed foods and beverages over a specific period, typically from the past few months to a year. FFOQs are cost-effective, suitable for large-scale studies, and provide insights into long-term dietary patterns. However, they may be subject to recall bias and do not capture detailed food preparation methods. Typically, FFOQs are developed and validated within a specific population. However, they can be modified to improve their applicability across cultures and regions. For example, the European Prospective Investigation into Cancer and Nutrition FFOQ, originally developed for the UK population, has been adapted for utilization in the Ukrainian population.¹⁷ From the United States, 2 major FFOQs used for large epidemiological studies are the National Cancer Institute (aka Block) and the Harvard (Willett) FFOQ.¹⁸ In addition, there has been interest in creating shorter questionnaires for assessment of adherence to the

DASH diet, including the DASH Online Questionnaire,¹⁹ DASH-Questionnaire,²⁰ and the DASH diet screener.²¹ Combining objective measures with self-reported dietary data can improve accuracy and personalization, though few validated biomarkers exist beyond single nutrients or foods.²² Although interest in dietary pattern biomarkers is growing, routine clinical use remains limited. Image-based food recognition systems, which use mobile cameras and computer vision to track and analyze food intake, are being actively developed to enhance dietary monitoring accuracy and efficiency while reducing the burden of traditional methods like food records or recalls.²³ A systematic review found that image-based food recognition systems demonstrated accuracy comparable to established methods when estimating energy content and food volumes, but showed lower accuracy in mixed or culturally diverse dishes.²⁴ Progress is still limited by a lack of standardization in food databases used to train artificial intelligence and reliance on food composition data that can be outdated, borrowed from other countries, or based on approximations of similar foods.²⁴ Furthermore, although some systems are described as fully automated, they typically still require human input during development, training, and sometimes in real-world applications to validate outputs or correct errors. Integrating novel technologies into Medical Nutrition Therapy dietitian consultations has significantly enhanced the precision, engagement, and effectiveness of personalized nutrition advice. These advancements enable dietitians to provide more tailored and impactful nutritional guidance with more informative and accurate data. For instance, some mobile apps have embedded

Table 1. Comparison of Commonly Utilized Dietary Assessment Methods

Method	Description	Advantages	Disadvantages
Weighed food records	Participants weigh all food and beverages consumed over a specific period, typically using a kitchen scale.	Highly accurate due to precise portion measurement.	Burdensome for participants; it can lead to participant drop-out.
		Reduces the risk of over- or underestimation of intake.	It can disrupt normal eating habits (participants choose items that are easier to weigh).
Food records (aka food diary)	Participants record all foods and beverages consumed over a specific time period, estimating portion sizes without weighing.	Easier and quicker for participants compared with weighed food records.	Estimations of portion sizes may be inaccurate.
		Less likely to alter normal eating behavior.	Participants may forget to record items or misreport intake.
		It can capture day-to-day variability if recorded over several days.	
Food recall	Participants are asked to recall all food and beverages consumed during the previous 24 h in detail. This may be led by an interviewer or automated tool (eg, intake24). The recall may involve prompts to aid memory (eg, using the multiple-pass method).	Minimal burden for participants as it does not require real-time recording.	Subject to memory and recall bias, especially for infrequent or snack foods.
		Can provide detailed information about food types, preparation methods, and nutrient intake.	Relies heavily on the participant's ability to estimate portion sizes accurately.
		Suitable for capturing data on populations where dietary habits vary day to day.	Single recalls may not accurately represent usual dietary intake; repeated recalls may be necessary.
FFQ	Participants are asked to recall all food and beverages consumed during a previous period of time (eg, past month or past year), usually through an interview or automated system, usually validated for a population.	Minimal burden for participants as it does not require real-time recording.	High reliance on participants' ability to remember and describe portion sizes.
		It can provide detailed information about food and	Subject to memory and recall bias.
		Nutrient intake.	
Image-based food recognition systems	Participants take photographs of their meals and snacks, which are then analyzed using software to estimate food intake and portion sizes.	Less effort for participants as it only requires taking photos.	It may miss details about ingredients or food preparation methods.
		It captures more accurate information on portion sizes with minimal participant input.	Technology may not always accurately identify portion sizes or mixed dishes (eg, the software may identify single foods such as an apple but have trouble with mixed foods such as curries).
		Real-time data collection reduces memory-related biases.	

FFQ indicates food frequency questionnaire.

image-based food recognition systems or guided diet diaries to facilitate data collection.²⁵ Reducing the time required for assessment and analysis of the data can increase the time available for nutrition education and appropriate interventions aligned with patients' health goals. Guiding principles have been developed so registered dietitians can apply personalized nutrition safely and clinically effectively.²⁶ Dietitians must be trained to interpret genetic, microbiome, and lifestyle data to tailor nutrient recommendations, while clearly communicating the benefits and limitations of personalized nutrition to support informed client engagement.

KEY FINDINGS AND LESSONS FROM PERSONALIZED NUTRITION AND GLYCEMIC RESPONSE

Inter-individual biological and lifestyle variability has been extensively studied to inform personalized nutrition approaches, with research focusing predominantly on average daily glucose concentrations and postprandial glycemic response, utilizing real-time blood glucose data generated by continuous glucose monitors.²⁷ Applications such as these have yet to be fully implemented to

address hypertension. Metabolic studies, where the primary outcome is independent of BP, represent a missed opportunity. Adding BP measurements utilizing gold-standard 24-hour ambulatory BP monitoring to these studies would significantly advance hypertension management.²⁸ Below, we summarize the key findings and lessons from personalized nutrition in glycemia studies and apply them to the future of personalized nutrition in hypertension.

One of the earliest studies demonstrated that genetically tailored diets, in which participants received personalized advice based on testing for 24 genetic variants across 19 genes, including those related to folate metabolism, antioxidant capacity, and blood glucose regulation, resulted in better compliance and long-term improvements in blood glucose levels compared with a standard Mediterranean diet.²⁹ A key benefit of specific recommendations is an improved impact on behavior change and adherence. Individuals respond better to personally tailored advice that feels more relevant to their specific lifestyle and biological design.³⁰ This was demonstrated in a randomized controlled trial (RCT) that observed a nutrigenomics-guided lifestyle intervention improved long-term dietary adherence compared with a standard weight management program.³¹

A cohort study of over 800 individuals revealed significant variability in postprandial glucose responses (PPGR) using a machine learning algorithm that incorporated factors like diet, activity, anthropometry, glucose levels, and gut microbiota.²⁷ Participants received short-term personalized diets through either an expert-designed approach or an algorithm-driven prediction arm, which ranked meals by PPGR and generated healthy and unhealthy plans.²⁷ The algorithm's effectiveness in reducing PPGR was comparable to expert intervention, demonstrating the potential of predictive models in personalized nutrition.²⁷ This algorithm was utilized in a single-center crossover pilot study ($n=23$) comparing personalized nutrition with a Mediterranean diet in newly diagnosed patients with type-2 diabetes, finding lower PPGR after 2 weeks of personalized intervention compared with the Mediterranean diet intervention (-19.8 ± 16.3 mg/dL $\times$ h, $P<0.001$).³² A subsequent 6-month follow-up showed significant improvements in HbA1c (glycated hemoglobin; $-0.39 \pm 0.48\%$, $P<0.001$), fasting insulin (-2.3 ± 4.0 MCU/mL, $P=0.04$), and triglycerides (-49 ± 46 mg/dL, $P<0.001$).³²

The PREDICT (Personalized Responses to Dietary Composition Trial) studies investigated how biological and personal factors influence individual responses to food.³³ The PREDICT-1 identified meal composition, genetics, meal context, glycemic markers, and gut microbiome as key contributors to PPGR variability.³³ Building on this, an 18-week RCT involving 225 participants showed that a personalized nutrition program delivered via a digital app significantly improved weight loss when compared with the provision of standard dietary guideline advice (mean difference, -2.51 kg [95% CI, -3.79 to -1.23]; $P<0.001$) and triglyceride levels (mean difference, -0.17 mmol/L [95% CI, -0.07 to -0.01]; $P=0.032$).³⁴ A 26-week study on individuals with impaired glucose regulation used DNA-personalized dietary advice based on 12 SNPs across 9 metabolism-related genes, including *FTO*, *APOC1*, and *PCSK9*.³⁵ Participants received either standard care as per National Institute for Health and Care Excellence guidelines (control arm), advice delivered by dietitians (intervention arm), or via a self-guided app and wearable device (exploratory arm).³⁵ Compared with the control arm, both personalized arms showed significant reductions in fasting glucose (intervention arm: -0.019 ± 0.008 , $P=0.01$; exploratory arm: -0.021 ± 0.008 , $P=0.006$) while only the dietitian-delivered intervention reduced HbA1c (intervention arm: -0.038 ± 0.018 , $P=0.05$; exploratory arm: $P>0.05$).³⁵

ROLE OF TECHNOLOGY IN PERSONALIZED NUTRITION DELIVERY

The development of continuous glucose monitors marked a pivotal technological breakthrough that

enabled real-time tracking of individual glycemic responses, generating vast data sets to train algorithms and accelerate personalized nutrition strategies for optimizing metabolic health. However, BP is measured infrequently for most people, with only a solitary reading commonly measured.³⁶ Currently, the gold standard measurement of BP remains ambulatory BP monitoring. This has resulted in a slower advance in personalized nutrition related to hypertension management, relative to progress regarding individual PPGRs, as ambulatory BP monitoring is not routinely performed on the same individual on multiple occasions. Over recent years, significant progress has been made in the development of cuffless BP measurement devices, which aim to provide continuous data streams comparable to those generated by continuous glucose monitor systems.³⁷ However, although these technologies are advancing, there remain notable challenges—particularly with measurement accuracy during overnight periods³⁸—and thus use in clinical practice remains limited.³⁷ Looking ahead, integrating emerging BP monitoring technologies with robust data from controlled dietary interventions or detailed dietary intake records may offer promising avenues for advancing personalized nutrition in hypertension management, provided the accuracy and reliability of these devices are sufficiently validated.

PERSONALIZED NUTRITION FOR HYPERTENSION

Personalized nutrition in the context of BP management has shown progress, particularly in relation to diet, genomics, and the gut microbiome (Figure 2). Due to rapid technological change in these areas, they offer tremendous opportunities to prevent and treat high BP. Indeed, a recent systematic review and meta-analysis demonstrated the efficacy of personalized nutrition interventions for decreasing SBP and diastolic BP (DBP).³⁹ The breadth of studies informing personalized nutrition approaches to hypertension, across diet, genomics, microbiome and metabolomics, is summarized in Table 2. Below, we highlight the key findings and gaps in the literature to accelerate this process from discovery to implementation.

Diet is one of the key lifestyle aspects of BP management.⁵⁵ Key recommendations include avoiding high-sodium foods and salty condiments, not adding salt in cooking or at the table, and consuming a diet rich in whole grains, fruits, and vegetables, which are high in potassium and fiber.⁵⁵ The recommendations for salt reduction are supported by extensive evidence from human clinical trials demonstrating that reducing sodium intake lowers BP.^{48,56–58} DASH or Mediterranean diets are examples of high potassium and high fiber diets that lower BP and are regularly recommended for

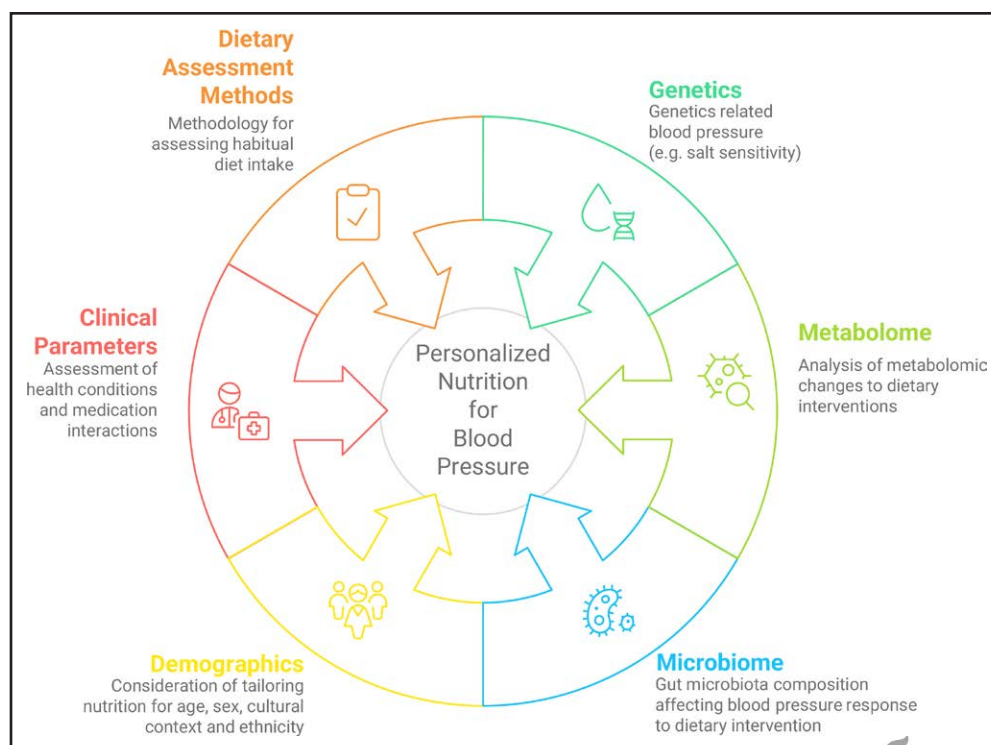


Figure 2. An overview of the factors contributing to the application of personalized nutrition for blood pressure (BP) management.

Personalized nutrition for BP management requires appropriate measurement of dietary intake and tailoring of subsequent dietary recommendations to an individual's unique characteristics, including genetics, microbiome composition, metabolic responses, and clinical characteristics. Schematic, not including text, generated using Napkin AI.

hypertension management.^{59,60} In addition, the modified DASH diet includes sodium restriction,⁶¹ which the Mediterranean diet does not, and can facilitate weight loss when total energy intake is reduced,⁶² which can lead to reductions in BP.⁶³ A recent umbrella review identified that, out of all the dietary patterns assessed, the DASH diet was the most effective for reducing BP, with high-quality evidence from RCTs.⁶⁴ A limitation is that these diets are more targeted at Westernized and European lifestyles and food availability.^{44,50} Economic barriers also limit adherence to the DASH, as it can be relatively more expensive.⁶⁵

Most global hypertension guidelines recommend limiting sodium intake for BP management, and although several advise increasing potassium intake, few offer specific recommendations regarding dietary potassium intake targets, despite its important role in BP control.⁶⁶ A 2020 meta-analysis of 32 RCTs found potassium supplementation lowers BP in a dose-dependent manner, especially in hypertensive individuals and those with high sodium intake, with a U-shaped relationship between potassium excretion and BP.⁶⁷ Potassium supplemented groups in the 32 RCTs had a mean difference of -3.9 mm Hg for SBP (95% CI, -5.2 to -2.6) and -2.4 mm Hg for DBP (95% CI, -3.8 to -1.1).⁶⁷ Typically, BP decrease with potassium supplementation was more effective in participants who were not on antihypertensive

medication, and in those who had the highest sodium exposure.⁶⁷ In the SSaSS (Salt Substitute and Stroke Study), an open-label, cluster-randomized trial of 20995 individuals from 600 villages in China, the difference in SBP between the potassium-enriched salt-substitute group and regular salt group was -3.34 mm Hg (95% CI, -4.51 to -2.18).⁶⁸ This trial also demonstrated a significant reduction in the primary outcome of stroke events (29.14 versus 33.65 events per 1000 person-years; rate ratio, 0.86 [95% CI, 0.77–0.96]; $P=0.006$) and the secondary outcome of major adverse cardiovascular events (49.1 versus 56.3 events per 1000 person-years; rate ratio, 0.87 [95% CI, 0.80–0.94]; $P<0.001$) in the salt-substitute group compared with the regular salt group.⁶⁸ Potassium supplementation can help lower BP by counteracting sodium's effects, but excessive intake—especially in patients with impaired kidney function—can cause harmful hyperkalemia.⁶⁹ In the SSaSS trial, 313 participants (302 deaths and 11 nonfatal events) had possible hyperkalemia, with no significant difference between the 2 trial arms (3.35 versus 3.30 events per 1000 person-years; rate ratio, 1.04 [95% CI, 0.80–1.37]; $P=0.76$).⁶⁸ Potassium-enriched salt substitutes have been recommended in recent guidelines for the management of hypertension,^{28,70} with meta-analysis demonstrating that they reduce SBP by 4.61 mm Hg ([95% CI, -6.07 to -3.14]; $P<0.001$), DBP by 1.61 mm Hg ([95% CI, -2.42

Table 2. Summary of Key Studies of Personalized Nutrition in Hypertension

Study reference	Personalization modality	Design	Country	Sample size	Population (Cohort)	Control	Intervention/ Comparator	BP measurement method	Main outcome
Appel et al ⁴⁰	Diet modification	Multicenter, randomized controlled feeding study	United States	n=459	Individuals with untreated hypertension, with DBP 80-95 mm Hg and SBP <160 mm Hg, with 2-thirds of minority background.	Typical American diet	Fruits-and-vegetables diet; combination diet rich in fruits, vegetables, and low-fat dairy foods	Office, ABPM	↓BP
Appel et al ⁴¹ and Loo et al ⁴²	Diet, metabolomics	Randomized, controlled, 3-period crossover feeding study	Baltimore, MD; Boston, MA	n=159–164	Individuals with untreated prehypertension and stage 1 hypertension (Omni-Heart Randomized Trial)	Protein and unsaturated fats diets	Carbohydrate diet	Office	↓BP ↓Triglycerides
Zeevi et al ²⁷	Machine learning-based algorithm predicting postprandial glyce-mic responses; microbiome metagenomics	Two-arm blinded randomized controlled trial	Israel	n=26	Nondiabetic adults	Postprandial glucose response from different diets as chosen by a clinical dietitian and researcher, as informed through continuous glucose monitoring	Postprandial glucose response from different diets as chosen by algorithm 	Office	Machine learning algorithm diets were comparable to RD diets
Aljuraiban et al ⁴³	Intake of dietary fiber variations in relation to BP	Cross-sectional study	United States	n=4680	US cohort of the INTERMAP study	BP in relation to low intake of dietary fiber in people with/without hypertension	BP in relation to high intake of dietary fiber in people with/without hypertension	Office	↓SBP
Kawamura et al ⁴⁴	Alteration to regional cuisines	Single-arm prospective study	Japan	n=58	Japanese individuals from Yamaguchi prefecture, Japan (DASH-JUMP trial)	Postintervention local diet	Localized DASH-modified diet	Home and office	↓BP ↓BMI ↓Fasting serum glucose level ↓Fasting insulin level
Baker et al ⁴⁵	Community-based nutrition education and improved access to heart-healthy opportunities.	Cross-sectional cohort study	Rural US	n=391–397	Black individuals from Pemiscot and Dunklin counties in Missouri (MOT-MGC)	Participants from Dunklin County.	Participants from Pemiscot County	Office	↓Prevalence in overweight and obese respondents ↓Prevalence of hypertension
Nierenberg et al ⁴⁶	BP genetic risk scores in salt-sensitive individuals	Single-arm family feeding study	Rural north China	n=1836–1850	Individuals from rural regions of Hebei, Henan, Shandong, Shaanxi and Jiangsu provinces of China (Gen-Salt study)	BP of individuals who are not salt/potassium sensitive	BP of salt/potassium sensitive individuals	Office	BP GRS quartiles were inversely proportional to BP response from salt and potassium intake.
Juraschek et al ⁴⁷	Sodium reduction and DASH diet effects on BP	Multicenter, parallel group, randomized crossover trial	United States	n=412	Healthy participants with SBP 120–159 mm Hg and DBP 80–95 mm Hg (DASH-Sodium trial)	Control diet and low-sodium level	DASH diet and medium/high-sodium levels	Office	↓SBP

(Continued)

Table 2. Continued

Study reference	Personalization modality	Design	Country	Sample size	Population (Cohort)	Control	Intervention/ Comparator	BP measurement method	Main outcome
Mill et al ⁴⁸	Salt-sensitivity across sex	Cross-sectional	Brazil	n=12 813	Civil servants of 6 Brazilian public universities or research institutions (ELSA-Brasil study)	Male individuals	Female individuals	Office	↑BP response in women
Berry et al ⁵³	Machine learning-based algorithm predicting postprandial glycemic responses	Single-arm mechanistic intervention study	United Kingdom and United States	n=1002 (United Kingdom) n=100 (United States)	Healthy adult individuals (ZOE PREDICT1 trial)	Postprandial metabolic responses at baseline	Postprandial metabolic responses after taking standardized test meals at clinic and home	Office	Large intraindividual variability was found in postprandial responses among participants.
Huart et al ⁴⁹	Metabolomics	Prospective, longitudinal study	Belgium	n=42	Nonhypertensive male and female volunteers across 3 cohorts	Nonhypertensive individuals in the cohorts	Hypertensive individuals in the cohorts	ABPM	↑Fecal acetate, propionate, butyrate
Wang et al ⁵⁰	Alteration to regional cuisines	Multicenter randomized feeding trial	China	n=120 (control) n=119 (CHH diet)	Chinese nationals from chosen Chinese regions (CHH trial)	Local diets as control	Cuisine-specific CHH diet	Office	↓SBP
Reay et al ⁵¹	Genetics to predict individual BP responses to sodium and potassium	Cross-sectional	Australia and United Kingdom	n=3207 (HCS); n=296 475 (UKBB)	HCS and UKBB cohort	Polygenic risk scores for hypertension	Pharmagenic Enrichment Scores for sodium and potassium pathways	Office	Urinary sodium and potassium could be used as proxies for salt-sensitivity.
Bermingham et al ⁵⁴	Dietary advice provided through an app, using a machine learning-based algorithm	Parallel design, randomized controlled trial	United States	n=170 (control) n=177 (intervention group)	Adult Americans reflective of the population (ZOE METHOD trial)	Generalized advice	Personalized dietary advice based on app algorithm	Office	↓Mean energy intake ↓Body weight ↓Waist circumference
Kallapura et al ⁵²	Gut microbiota-based personalized nutrition	Randomized prospective trial	Bhopal, India	n=30	Hyperglycemic and hyperlipidemic individuals from Bhopal, India	Regular diet	BugSpeaks gut microbiome-based personalized diet	Office	↓HbA1c ↓SBP ↑Microbiome diversity
Verhaar et al ⁵³	Effects of oral butyrate on BP	Double-blind, randomized, placebo-controlled trial	Amsterdam, the Netherlands	n=21	Individuals with untreated hypertension	Placebo capsules with sodium chloride	Oral sodium butyrate capsules	ABPM	↑Daytime 24 h BP
Sun et al ⁵⁴	Metabolomics	Phase III randomized, double-blind, crossover, placebo-controlled trial	New Orleans, LA; Jackson, MS	n=80 (ProBP trial) n=1131 (BHS trial)	Individuals with untreated prehypertension and stage1 hypertension (ProBP trial)	Carbohydrate diet	Soy protein and milk protein supplementation	Office	Eight metabolites, including 7 novel, were significantly associated with BP.

ABPM indicates ambulatory blood pressure monitoring; BMI, body mass index; BP, blood pressure; CHH, Chinese Heart-Healthy; DASH, Dietary Approaches to Stop Hypertension; DASH-JUMP, DASH–Japan Ube Modified Diet Program; DBP, diastolic blood pressure; GRS, genetic risk scores; HbA1c, glycated hemoglobin; HCS, Hunter Community Study; INTERMAP, International Study of Macro/Micronutrients and Blood Pressure; ProBP, Protein and Blood Pressure; RD, registered dietitian; SBP, systolic blood pressure; and UKBB, UK Biobank.

to -0.79]; $P<0.001$), and overall cardiovascular risk (risk ratio, 0.87 [95% CI, 0.81 – 0.94]; $P<0.001$).⁷¹ However, these clinical benefits were predominantly driven by trials conducted in East Asian populations, most notably the SSaSS trial in rural China, and these results may not be globally generalizable to populations where dietary

sodium is largely derived from processed foods rather than discretionary salt use.⁷¹

The principles of these BP-lowering diets can be personalized and applied to more diverse cuisines and ethnicities to tackle hypertension in various cultural backgrounds.^{44,45,50} An RCT in 265 Chinese participants

showed the Chinese Heart-Healthy diet significantly reduced SBP by 10 mmHg ([95% CI, −12.1 to −7.9]; $P<0.001$) and DBP by 3.8 mmHg ([95% CI, −5.0 to −2.5]; $P<0.001$) compared with the control group.⁵⁰ The diet was based on the Chinese Dietary Guidelines, DASH principles, and national intake data, with an added focus on affordability. It achieved a 1 mmHg SBP reduction for just 0.4 CNY (0.06 USD) per day, with participants reporting that the palatability of the intervention diet was similar to their habitual diet.⁵⁰ The DASH-JUMP trial (DASH–Japan Ube Modified Diet Program) was an open-label single-arm trial involving 58 participants with stage 1 hypertension who underwent a 2-month dietary intervention with a DASH diet tailored to Japanese tastes.⁴⁴ After this 2-month intervention, there were significant reductions from baseline in both SBP (baseline: 153±14 mmol/L versus 2 months: 130±16 mmol/L, $P<0.01$) and DPB (baseline: 91±11 mmol/L versus 2 months: 80±9 mmol/L, $P<0.01$) and at the 6-month time point, 4 months after the intervention finished, both SBP (139±16 mmol/L, $P<0.01$) and DBP (85±10 mmol/L, $P<0.01$) remained significantly lower compared with baseline.⁴⁴ These studies highlight how culturally adapted precision nutrition can effectively target specific populations to reduce BP.

GENOMICS

Genetics is a key component of personalized medicine, originating from the concept that individual genetic variations will result in differential disease risks and medication responses. Patients presenting rare monogenic (Mendelian) forms of hypertension, who usually present with large effects on BP (≈20–50 mmHg), have been described.⁷² These are usually due to pathogenic variants in sodium-handling genes⁷³ and are observed in younger individuals.⁷² Most hypertension cases are multifactorial, involving genetic and lifestyle factors, with ≈2100 SNPs identified by genome-wide association studies explaining over 60% of SNP-based heritability and 6.93% of SBP variance.⁷⁴ This has enabled the development of polygenic risk scores (PRS) to predict the risk of hypertension. Individuals in the top versus bottom PRS deciles exhibited up to a 7-fold difference in hypertension risk (odds ratio, 7.33 [95% CI, 5.54–9.70]; $P=4.13\times10^{-44}$) corresponding to a 16.9 mmHg difference in SBP ([95% CI, 15.5–18.2 mmHg]; $P=2.22\times10^{-126}$).⁷⁴ Our understanding of genetics in BP regulation has grown significantly, but gene–environment interactions—especially those involving diet—remain poorly understood and are key to advancing personalized nutrition, pharmacogenomics, and hypertension risk assessment.⁷⁵

Variability in BP response to nutrients has been associated with factors such as sex,⁷⁶ ethnicity,¹¹ health status,^{77,78} and socioeconomic status.⁷⁹ Personalized nutrition should be tailored to consider sex-based physiological

differences and socially constructed factors. Macronutrient metabolism in men and women differs, resulting in distinct body fat distribution driven by hormones. In men, body fat tends to be stored in the visceral region, which has been associated with increased risk of insulin resistance.⁸⁰ This can be assessed with the waist-hip ratio, which is also the best discriminator for cardiovascular risk factors, including hypertension, type-2 diabetes, and dyslipidemia.⁸¹ In the UK Biobank study, women were observed to have higher intakes of total fat, saturated fat, and total sugar compared with recommended targets.⁸² Although dietary fiber targets to reduce BP differ between males and females, recommended sodium targets are the same. However, more data are needed to support sex-specific personalized nutrition interventions for hypertension.

In addition, some individuals are genetically predisposed to have an exaggerated BP response to sodium intake, called salt-sensitivity, which puts these individuals at an elevated risk of cardiovascular events.⁸³ For instance, salt-sensitivity SNPs have been identified in *PTGS* (prostaglandin-endoperoxide synthase) and *PTGS2* genes, which are rate-limiting prostaglandin enzymes associated with injury, inflammation, and proliferation.⁸⁴ Six *PTGS2* variants were found. The SNP rs12042763 was positively associated with SBP response to a low-salt diet and negatively associated with SBP response to a high-salt diet.⁸⁴ Variants in the sodium bicarbonate cotransporter gene *SLC4A5*, specifically rs7571842 (odds ratio, 0.221, $P=1.04\times10^{-4}$) and rs10177833 (odds ratio, 0.221, $P=3.1\times10^{-4}$), were associated with salt-sensitivity in 2 independent European populations.⁸³ Salt-sensitivity has been reported more frequently among individuals of African ancestry,¹¹ and appears to be more pronounced among hypertensive, HIV-positive individuals in sub-Saharan Africa.⁷⁷ Furthermore, among HIV-positive individuals in sub-Saharan Africa, elevated immune activation has been associated with an increased risk of hypertension.⁷⁸ Assessment of salt-sensitivity is usually laborious and costly, with the reference method being a controlled dietary protocol of 1-week on a low-sodium diet (<50 mmol NaCl/d, or 2922 mg NaCl/d) followed by 1-week on a high-sodium (≈250 mmol NaCl/d, or 14610 mg NaCl/d) and thus, it is rarely performed.⁸⁵ Hence, nutrigenomics could potentially play an important role in identifying salt-sensitive individuals and reducing their associated disease burden.

Genomic PRS derived from hypertension and BP genome-wide association studies lacked precision in detecting potential sodium responses at individual levels. For example, in the GenSalt study, baseline BP was positively associated with BP PRS, but dietary intervention responses were inversely related.⁴⁶ When modified to known biological networks, PRS can identify responses to interventions, known as a Pharmagenic Enrichment Score (PES). Using the UK Biobank, PES originating

from these gene sets associated with nutrient absorption (64 unique genes), sodium and potassium transport (487 unique genes), sodium and potassium renal excretion (93 unique genes), and potassium-only transport (264 unique genes) were assessed in 296 475 participants.⁵¹ Participants with higher PES, particularly in sodium and potassium transport, received an outsized benefit on SBP in response to sodium reduction, proxied by urinary sodium, with each SD decrease in urinary sodium corresponding to a 1.47 mmHg decrease in BP (95% CI, 1.24–1.7) among participants with a top decile PES, compared with a 0.97 mmHg (95% CI, 0.75–1.19) increase for those with a PES in the lowest decile.⁵¹ In this context, urinary sodium was used as a proxy for dietary sodium intake. This potent gene-by-environment interaction persisted after adjusting for kidney function and urinary creatinine, and was not observed with genome-wide polygenic risk for BP.⁵¹ This approach could, therefore, be used to identify salt-sensitivity and target responders to sodium interventions, and strongly support further investigation in prospective intervention trials.

GUT MICROBIOME

It is increasingly recognized that the gut microbiota plays a critical role in modulating BP, with emerging evidence suggesting that microbial metabolites such as short-chain fatty acids (SCFA) and trimethylamine-*N*-oxide can influence BP.^{86,87} The gut microbiota also modulates individual responses to dietary interventions,⁸⁸ indicating that it may be a potent target for personalized nutritional advice to lower BP. More than 20% of interpersonal gut microbial variation has been attributed to environmental factors such as diet,⁸⁹ leading to highly personalized responses in diet-microbiome interactions.³³ A small trial profiled the gut microbiota of hyperglycemic and hyperlipidemic participants and provided personalized dietary recommendations for 3 months, with a nonsignificant decrease in SBP of 7 mmHg (5%) observed in the personalized nutrition intervention arm, while those receiving traditional dietary advice observed a 9 mmHg (7%) increase in SBP.⁵² In participants whose baseline SBP was >130 mmHg, the reduction in SBP from 153 mmHg (95% CI, 138.7–167.2) to 131 mmHg (95% CI, 125.8–137.7) was significant ($P<0.01$) at the end of the microbiome-based personalized dietary intervention.⁵²

Dietary fiber is a potent modulator of the gut microbiome, which has traditionally been classified based on water solubility, using a binary system of soluble versus insoluble fiber.⁹⁰ In the INTERMAP study (International Study of Macro/Micronutrients and Blood Pressure), an increase in total fiber intake by 6.8 g/4184 kJ (6.8g/1000 kcal) was associated with a 1.69 mmHg lower SBP (95% CI, –2.97 to –0.41).⁴³ An increased intake of insoluble fiber by 4.6 g/4184 kJ (4.6 g/1000 kcal) was associated with a 2.07 mmHg reduction in

SBP (95% CI, –3.92 to –0.23) after adjustment for lifestyle factors, body mass index, and soluble fiber intake. However, this association did not remain statistically significant after adjustment for urinary potassium levels.⁴³ No significant association was observed for soluble fiber intake.⁴³ These findings are consistent with a meta-analysis evaluating the antihypertensive effects of 5 types of viscous soluble fibers (β -glucan, guar gum, konjac, pectin, and psyllium), which found no significant BP-lowering effects except for a modest reduction in SBP with psyllium (mean difference, –2.39 mmHg [95% CI, –4.62 to –0.17]).⁹¹ However, categorizing fibers solely as soluble or insoluble fails to capture their full physiological functionality. Characteristics such as viscosity, fermentability, and bulking capacity play critical roles in mediating their health effects.⁹⁰ Many dietary fibers exhibit multiple functional properties; for example, β -glucans—found in oats and known for their beneficial role in reducing cholesterol reabsorption—are soluble and moderately fermentable, yet contribute minimally to stool bulk. Understanding the health effects of dietary fiber requires consideration of its functional properties beyond solubility alone.

There is accumulating evidence that fermentable fiber exerts BP-lowering effects via the microbial production of SCFA, primarily acetate, propionate, and butyrate (Figure 3).^{92–97} Evidence increasingly points to fiber fermenters and SCFA-producing taxa as a key indicator of a healthy gut microbiome.⁹⁸ SCFA producers such as *Faecalibacterium prausnitzii* and *Roseburia hominis* are negatively correlated with BP and depleted in patients with hypertension.^{99–101} An individual's baseline dietary fiber intake influences the microbial response to dietary fiber interventions, in a manner that is distinct between different interventions.¹⁰² Incorporating different forms of fiber and acetate-containing products such as apple cider vinegar boosts fecal SCFA concentrations and colonic fermentation of fiber.¹⁰³ Although oral direct delivery of SCFA was promising in hypertensive animal models,^{92,93} this method is not feasible in humans.¹⁰⁴ A crossover RCT ($n=20$) utilizing acetylated and butyrylated starches, providing colonic delivery of SCFA, reported antihypertensive effects from baseline to end point (daytime SBP: treatment arm 142.1 ± 10.6 – 138.1 ± 10.3 mmHg; placebo arm 139.7 ± 11.6 – 140.2 ± 10.8 mmHg).⁹⁵ However, a second crossover RCT utilizing 4 weeks of supplementation with sodium butyrate capsules found it to be ineffective, with reports of increased daytime SBP from baseline to end point in the butyrate intervention group (134.6 ± 12.6 – 144.7 ± 11.1 mmHg) while it remained unchanged in the placebo group (138.1 ± 12.9 – 138.6 ± 12.3 mmHg).⁵³ The discordant results observed between these studies suggest that the beneficial effects of SCFA supplementation may depend on delivery to the colonic region, rather than the administration of free SCFAs, which are rapidly absorbed in the small intestine.¹⁰⁵ Indeed, the colon is the primary site inhabited by SCFA-producing gut bacteria.

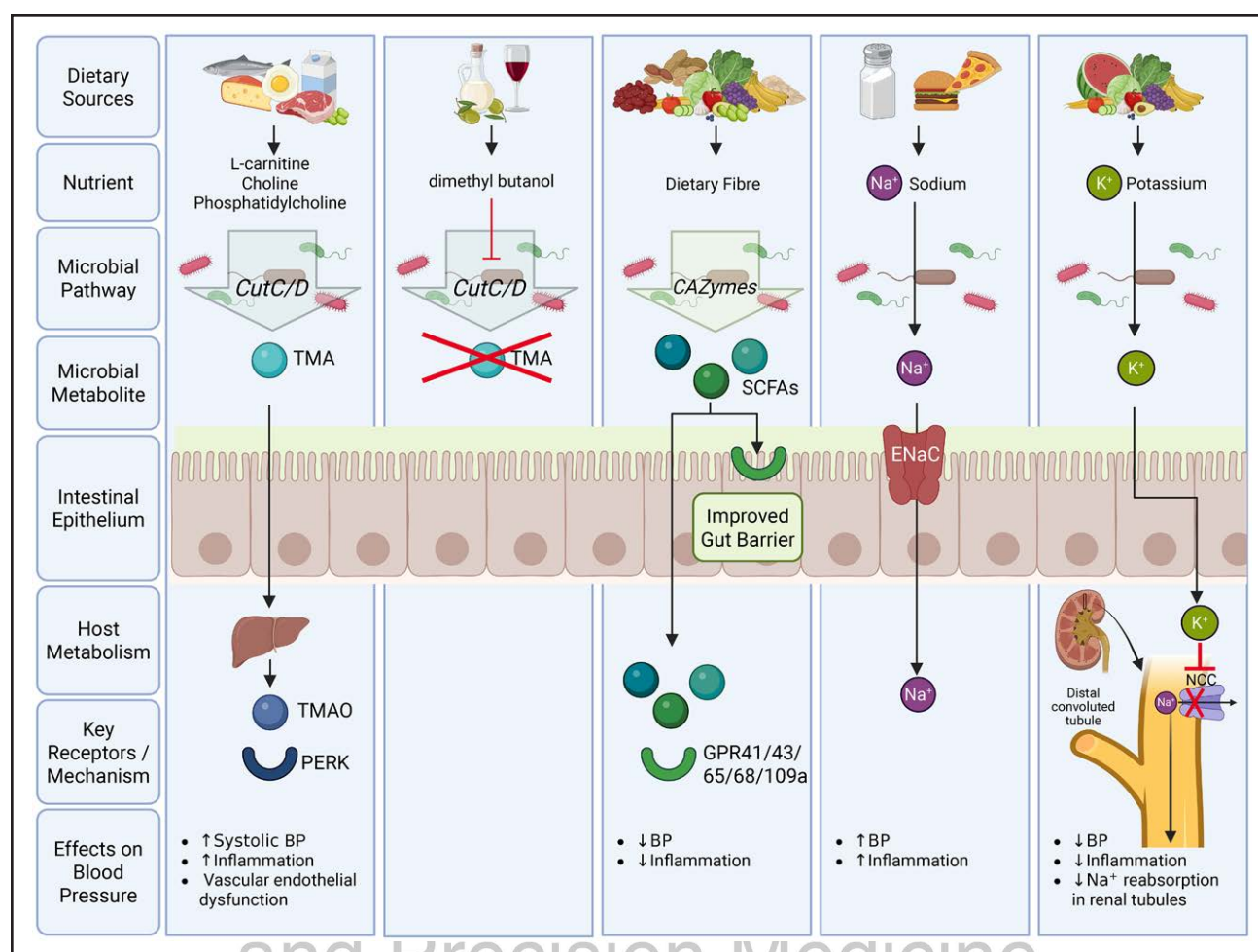


Figure 3. Dietary pathways influencing blood pressure (BP).

Dietary components shape gut microbial metabolism and downstream host signaling pathways that regulate BP and vascular inflammation. CAZymes indicates carbohydrate-active enzymes; CutC/CutD, choline trimethylamine lyase enzymes C and D; ENaC, epithelial sodium channel; GPR, G-protein-coupled receptor; NCC, Na⁺/Cl-cotransporter; PERK, protein kinase R-like endoplasmic reticulum kinase; SCFA, short-chain fatty acids; TMA, trimethylamine; and TMAO, trimethylamine-*N*-oxide. Created using BioRender.

Therefore, enhancing SCFA delivery to this region may improve activation of endogenous SCFA-mediated pathways,¹⁰⁶ including an improvement in gut epithelial barrier.¹⁰⁷ Further research is needed to understand underlying interactions that elicit varied responses in individuals, particularly due to limited longitudinal intervention studies on indigenous populations, which may still possess human microbial profiles lost to the modern populations.¹⁰⁸

Another consideration is gene-by-environment interactions, whereby the host genetics are responsible for sensing SCFA and facilitating their BP-lowering effects. This is primarily due to activation of SCFA receptors, the GPR (G-protein-coupled receptor) 41 (*FFAR3*) and GPR43 (*FFAR2*). Using animal models and human genetics, it became clear that the lack of GPR41/43 leads to an increase in BP, and GPR41/43 are (at least partially) responsible for mediating fiber's BP-lowering effects.^{93,109} Population genetics data in ≈300 000 UK

Biobank participants show that individuals with higher *FFAR2/3* expression (estimated using expression quantitative trait locus) have a lower prevalence of hypertension (37.3% homozygous carriers of both alleles versus 38% in the rest of the cohort, $P=0.00763$).¹¹⁰ Indeed, people with hypertension have significantly lower levels of *FFAR2* mRNA in circulating white blood cells ($P<0.001$), even after adjusting for age, body mass index and sex ($\beta=-12.3$, $P=0.004$).¹⁰⁹ Moreover, in ≈400 000 participants, carriers of rare pathogenic variants in these genes, which lead to the absence of their protein, have a higher odds ratio for hypertension and major adverse cardiovascular events, even if their fiber intake is adequate.¹¹¹

METABOLOMICS

The metabolome provides a dynamic snapshot of physiological state, reflecting responses to internal and external conditions; metabolites serve as biomarkers for early

disease detection, monitoring, treatment, and potentially dietary intake.¹¹² The OmniHeart study randomized 164 participants in a crossover feeding study to receive diets rich in carbohydrate, protein, or monounsaturated fat.⁴¹ A follow-up analysis of the OmniHeart study used urinary proline-betaine to assess diet adherence, finding hippurate positively and 4-cresyl sulfate and phenylalanine negatively associated with BP in adherent participants.⁴² While dietary interventions are often prescribed, adherence is uncertain; metabolomics offers an objective tool to assess compliance and its impact on health outcomes.

Metabolites linked to BP vary by dietary intervention, with little overlap across studies, and feeding trials highlight individual variability in response to the same diet.^{54,113,114} In a secondary analysis of the ProBP trial (Protein and Blood Pressure; $n=80$), plasma metabolites were associated with BP after interventions with carbohydrates, soy protein or milk protein, with several of these metabolites validated in a larger cross-sectional study.⁵⁴ *N*-stearoylsphinganine (d18:0/18:0), a ceramide that mediates inflammatory responses, was one such metabolite⁵⁴ that was previously observed to be reduced after a DASH diet intervention.¹¹⁵ Higher plasma concentrations of a different ceramide (d18:1,C24:0) were associated with BP in a Swedish cohort study; however, this was not confirmed in a validation cohort.¹¹³ Conversely, in a subset from the GenSalt replication study, 161 urinary metabolites differed between 50 salt-sensitive and 50 salt-resistant participants, but none remained significant after correcting for multiple testing.¹¹⁴ Substantial variability exists in plasma and urinary metabolomics results for identifying consistent BP biomarkers.

Increasingly, the fecal metabolome is being analyzed to provide insights into the modulation and perturbations of the gut microbiota in hypertension and to assess the impact of dietary interventions on metabolites produced by the gut microbial consortium.^{92,95} Known metabolites associated with diet and hypertension include trimethylamine-*N*-oxide,¹¹⁶ SCFA,⁴⁹ indole-3-acetic acid, and indole-3-lactic acid.¹¹⁷ Trimethylamine, the precursor to trimethylamine-*N*-oxide, is produced from choline by microbial CutC and CutD (choline trimethylamine lyase enzymes C and D) proteins, which cleave choline for trimethylamine synthesis.¹¹⁸ The choline analogues dimethyl butanol,¹¹⁹ iodomethylcholine, and fluoromethylcholine, which inhibit trimethylamine-*N*-oxide production via CutC and CutD proteins, are currently being investigated in preclinical models, with promising results.¹¹⁸ In addition, dimethyl butanol naturally occurs in fermented beverages and oils such as extra virgin olive oils, grape seed oils, and red wines, therefore making it an ideal candidate in dietary interventions involving gut microbial trimethylamine production.¹¹⁹ The male offspring of female rats that received dimethyl butanol orally were protected from hypertension induced by environmental exposure.¹²⁰ Although an emerging area, the fecal

metabolome is revealing key microbial metabolites linked to BP and uncovering novel therapeutic approaches.

LIMITATIONS AND FUTURE DIRECTIONS

Personalized nutrition drives better BP management in individuals compared with standard guidelines for the public, as evidenced in a meta-analysis of 15 studies, with SBP, weighted-mean differences decreasing by -1.91 (95% CI, -3.51 to -0.31 mm Hg) and diastolic BP by -1.49 (95% CI, -2.39 to 0.58 mm Hg) relative to standard nutrition guidelines.³⁹ Despite its potential in managing BP, personalized nutrition faces several barriers to widespread adoption. Genomic and microbiome research is disproportionately concentrated in high-income countries, leaving low- and middle-income countries underrepresented—for example, India, Pakistan, and Bangladesh contribute only 1.8% of global microbiome samples despite comprising 25% of the population.¹²¹ Low- and middle-income countries may have limited resources and access to technology due to various socioeconomic, financial, and political factors, which hinder implementation in personalized nutrition.¹²² Although high-income countries benefit from greater feasibility, discrepancies in training, confidence, and knowledge of genetics and nutritional genomics in dietitians have been reported.^{123,124} Within high-income countries, access to personalized nutrition is not homogeneous across the nation, as diet-related health inequalities arise from factors such as ethnicity, occupation, religion, and socioeconomic status.¹²⁵ Even in high-income countries such as the United States, it has been observed that there are lower implementation rates of genomics testing in racial minority groups, rural communities, and lower-income individuals, which may further fuel health disparities for these groups.¹²⁶ Cost-effective, scalable models—such as the RESTORE Network in the United States—could guide adaptation for personalized nutrition.¹²⁷ Integration into healthcare systems, clinician training, culturally tailored interventions, and technology use are key to expanding access.²⁶ Multidisciplinary collaboration is essential to ensure equity and develop adaptable strategies across diverse populations. Emerging artificial intelligence tools, including machine learning models that predict hypertension risk and support personalized lifestyle guidance, offer promise but must be validated across populations to avoid reinforcing disparities.¹²⁸ In addition, personalized nutrition in managing hypertension is limited by heterogeneity of randomized clinical trials in terms of intervention length, frequency of follow-up, and a lack of reporting of subsequent physiological changes such as arterial stiffness, which contribute to the development of cardiovascular disease.³⁹ Longer interventions and follow-up periods of at least 12 months have shown higher adherence to dietary advice among participants and

exhibit long-term impacts on outcomes.^{39,129} In particular, for hypertension, there is a need for consistent use of 24-hour BP measurements for more reliable and comparable results.³⁹ Furthermore, the food industry, as a predominant source of processed foods containing unregulated quantities of salt, sugar, and insufficient dietary fiber and potassium, is a contributor to hypertension. National regulatory policies that engage with the food industry can help reduce the prevalence of hypertension, as evidenced by the National Salt Reduction Initiative, led by the New York City Health Department.¹³⁰ With these in place, personalized nutrition efforts can be augmented to achieve better outcomes.

CONCLUSIONS

Personalized nutrition represents a promising approach to addressing the global challenge of hypertension by tailoring dietary interventions based on an individual's unique biological and lifestyle factors. Integrating genomics, gut microbiome, and metabolic phenotype profiling into dietary recommendations has shown potential for more precise and effective BP management than traditional one-size-fits-all guidelines. Although current evidence highlights significant improvements in BP control through personalized dietary strategies, there are still limitations regarding accessibility, long-term effectiveness, clinical application and integration into healthcare settings, and benefit to a more diverse group of patients. Advances in technologies for continuous BP monitoring and computer-aided dietary assessment methods will provide essential data sources for machine learning models to provide personalized nutrition recommendations to manage hypertension. Developing cost-effective ways to implement personalized nutrition in diverse healthcare settings remains a key challenge. Despite these hurdles, personalized nutrition is poised to become an important component of precision medicine for hypertension management, offering a path towards more targeted and sustained reductions in BP.

ARTICLE INFORMATION

Affiliations

Department of Pharmacology, Biomedicine Discovery Institute, Faculty of Medicine, Nursing and Health Sciences, Monash University (P.C., C.M., F.Z.M., M.S.). Victorian Heart Institute, Monash University, Clayton, Australia (P.C., C.M., F.Z.M., M.S.). School of Health Sciences, College of Health, Medicine and Wellbeing (E.D.C., J.S., C.E.C.) and School of Biomedical Science & Pharmacy (M.J.C.), University of Newcastle, Callaghan, Australia. Food & Nutrition Research Program (E.D.C., J.S., C.E.C.) and Precision Medicine Research Program (M.J.C.), Hunter Medical Research Institute, New Lambton Heights, Australia. Baker Heart and Diabetes Institute, Melbourne, Australia (F.Z.M.). Nutrition & Health Innovation Research Institute, Edith Cowan University, Perth, Australia (M.S.).

Sources of Funding

Dr Marques is supported by a Senior Medical Research Fellowship from the Sylvia and Charles Viertel Charitable Foundation, a National Heart Foundation Future

Leader Fellowship (105663), and National Health & Medical Research Council (NHMRC) Emerging Leader Fellowship (GNT2017382). Dr Snelson is supported by a National Heart Foundation Postdoctoral Fellowship (106698). Dr Collins is supported by an NHMRC Leadership Investigator Grant (GNT2009340). Dr Cairns is supported by an NHMRC Leadership Investigator Grant (GNT2025656).

Disclosures

Dr Cairns is a director and shareholder of PolygenRx Pty Ltd, which is developing and commercializing the Pharmagenic Enrichment Score platform and associated intellectual property. The other authors report no conflicts.

REFERENCES

1. World Health Organization. Global report on hypertension: the race against a silent killer. 2023
2. Global Health Observatory. *Noncommunicable Diseases: Risk Factors*. World Health Organization; 2023.
3. Murray CJL, Aravkin AY, Zheng P, Abbafati C, Abbas KM, Abbasi-Kangevari M, Abd-Allah F, Abdelalim A, Abdollahi M, Abdollahpour I, et al. Global burden of 87 risk factors in 204 countries and territories, 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet*. 2020;396:1223–1249. doi: 10.1016/s0140-6736(20)30752-2
4. Muralitharan R R, Marques FZ, O'Donnell JA. Recent advancements in targeting the immune system to treat hypertension. *Eur J Pharmacol*. 2024;983:177008. doi: 10.1016/j.ejphar.2024.177008
5. Valenzuela PL, Carrera-Bastos P, Gálvez BG, Ruiz-Hurtado G, Ordovas JM, Rulope LM, Lucia A. Lifestyle interventions for the prevention and treatment of hypertension. *Nat Rev Cardiol*. 2021;18:251–275. doi: 10.1038/s41569-020-00437-9
6. Volpp KG, Berkowitz SA, Sharma SV, Anderson CAM, Brewer LC, Elkind MSV, Gardner CD, Gervis JE, Harrington RA, Herrero M, et al; American Heart Association. Food is medicine: a presidential advisory from the American Heart Association. *Circulation*. 2023;148:1417–1439. doi: 10.1161/CIR.0000000000001182
7. Senkus KE, Dudzik JM, Lennon SL, DellaValle DM, Moloney LM, Handu D, Rozga M. Medical nutrition therapy provided by a dietitian improves outcomes in adults with prehypertension or hypertension: a systematic review and meta-analysis. *Am J Clin Nutr*. 2024;119:1417–1442. doi: 10.1016/j.ajcnut.2024.04.012
8. Alameddine M, Nasreddine L, Hwalla N, Mourad Y, Shoaib H, Mousa D, Naja F. Factors associated with consulting a dietitian for diabetes management: a cross-sectional study. *BMC Health Serv Res*. 2013;13:504. doi: 10.1186/1472-6963-13-504
9. Bailey RL, Stover PJ. Precision nutrition: the hype is exceeding the science and evidentiary standards needed to inform public health recommendations for prevention of chronic disease. *Annu Rev Nutr*. 2023;43:385–407. doi: 10.1146/annurev-nutr-061021-025153
10. Masenga SK, Wandira N, Cattivelli-Murdoch G, Saleem M, Beasley H, Hinton A Jr, Ertuglu LA, Mwesigwa N, Kleyman TR, Kirabo A. Salt sensitivity of blood pressure: mechanisms and sex-specific differences. *Nat Rev Cardiol*. 2025;22:611–628. doi: 10.1038/s41569-025-01135-0
11. Eljovich F, Kirabo A, Laffer CL. Salt sensitivity of blood pressure in black people: the need to sort out ancestry versus epigenetic versus social determinants of its causation. *Hypertension*. 2024;81:456–467. doi: 10.1161/HYPERTENSIONAHA.123.17951
12. Wright JT, Rahman M, Scarpa A, Fatholahi M, Griffin V, Jean-Baptiste R, Islam M, Eissa M, White S, Douglas JG. Determinants of salt sensitivity in black and white normotensive and hypertensive women. *Hypertension*. 2003;42:1087–1092. doi: 10.1161/01.HYP.0000101687.89160.19
13. Galekop MMJ, Uyl-de Groot CA, Ken Redekop W. A systematic review of cost-effectiveness studies of interventions with a personalized nutrition component in adults. *Value Health*. 2021;24:325–335. doi: 10.1016/j.jval.2020.12.006
14. Kostova D, Spencer G, Moran AE, Cobb LK, Husain MJ, Datta BK, Matsushita K, Nugent R. The cost-effectiveness of hypertension management in low-income and middle-income countries: a review. *BMJ Glob Health*. 2020;5:e002213. doi: 10.1136/bmjgh-2019-002213
15. Naska A, Lagiou A, Lagiou P. Dietary assessment methods in epidemiological research: current state of the art and future prospects. *F1000Res*. 2017;6:926. doi: 10.12688/f1000research.10703.1
16. McCambridge J, Witton J, Elbourne DR. Systematic review of the Hawthorne effect: new concepts are needed to study research participation effects. *J Clin Epidemiol*. 2014;67:267–277. doi: 10.1016/j.jclinepi.2013.08.015

17. Shatylo S, Solovyova G. Adaptation and validation of the EPIC-Norfolk food frequency questionnaire for assessing dietary intake in Ukrainian adults. *BMJ Nutr Prev Health*. 2024;7:160–165. doi: 10.1136/bmjnp-2023-000703
18. Stark A. An historical review of the Harvard and the National Cancer Institute food frequency questionnaires: their similarities, differences, and their limitations in assessment of food intake. *Ecol Food Nutr*. 2002;41:35–74. doi: 10.1080/03670240212531
19. Apovian CM, Murphy MC, Cullum-Dugan D, Lin PH, Gilbert KM, Coffman G, Jenkins M, Bakun P, Tucker KL, Moore TJ. Validation of a web-based dietary questionnaire designed for the DASH (Dietary Approaches to Stop Hypertension) diet: the DASH online questionnaire. *Public Health Nutr*. 2010;13:615–622. doi: 10.1017/S1368980009991996
20. Warren-Findlow J, Reeve CL, Racine EF. Psychometric validation of a brief self-report measure of diet quality: the DASH-Q. *J Nutr Educ Behav*. 2017;49:92–99.e1. doi: 10.1016/j.jneb.2016.09.004
21. Yao Q, Cohen SA, Oaks BM, Vadiveloo MK. Brief Diet screener for assessing adherence to the Dietary Approaches to Stop Hypertension (DASH) diet: development and validation. *Curr Dev Nutr*. 2025;9:107593. doi: 10.1016/j.cdnut.2025.107593
22. Clarke ED, Rollo ME, Pezdirc K, Collins CE, Haslam RL. Urinary biomarkers of dietary intake: a review. *Nutr Rev*. 2020;78:364–381. doi: 10.1093/nutrit/nuz048
23. Dalakleidi KV, Papadelli M, Kaposos I, Papadimitriou K. Applying image-based food-recognition systems on dietary assessment: a systematic review. *Adv Nutr*. 2022;13:2590–2619. doi: 10.1093/advances/nmac078
24. Shonkoff E, Cara KC, Pei X, Chung M, Kamath S, Panetta K, Hennessy E. AI-based digital image dietary assessment methods compared to humans and ground truth: a systematic review. *Ann Med*. 2023;55:2273497. doi: 10.1080/07853890.2023.2273497
25. Crocombe D, Armes S, Anderson S, Johnsen J, Ray S. Empowering global nutrition with digital technology: Proceedings of the 8th annual International Summit on Nutrition and Health. *BMJ Specialist J*. 2023.
26. Adams SH, Anthony JC, Carvajal R, Chae L, Khoo CSH, Latulippe ME, Matusheski NV, McClung HL, Rozga M, Schmid CH, et al. Perspective: guiding principles for the implementation of personalized nutrition approaches that benefit health and function. *Adv Nutr*. 2020;11:25–34. doi: 10.1093/advances/nmz086
27. Zeevi D, Korem T, Zmora N, Israeli D, Rothschild D, Weinberger A, Ben-Yacov O, Lador D, Avnit-Sagi T, Lotan-Pompan M, et al. Personalized nutrition by prediction of glycemic responses. *Cell*. 2015;163:1079–1094. doi: 10.1016/j.cell.2015.11.001
28. McEvoy JW, McCarthy CP, Bruno RM, Brouwers S, Canavan MD, Cecconi C, Christodorescu RM, Daskalopoulou SS, Ferro CJ, Gerdtz E, et al; ESC Scientific Document Group. 2024 ESC guidelines for the management of elevated blood pressure and hypertension. *Eur Heart J*. 2024;45:3912–4018. doi: 10.1093/eurheartj/ehae178
29. Arkadianos I, Valdes AM, Marinou E, Florou A, Gill RD, Grimaldi KA. Improved weight management using genetic information to personalize a calorie controlled diet. *Nutr J*. 2007;6:1–8. doi: 10.1186/1475-2891-6-29
30. Dijksterhuis GB, Bouwman EP, Taufik D. Personalized nutrition advice: preferred ways of receiving information related to psychological characteristics. *Front Psychol*. 2021;12:575465. doi: 10.3389/fpsyg.2021.575465
31. Horne J, Gilliland J, O'Connor C, Seabrook J, Madill J. Enhanced long-term dietary change and adherence in a nutrigenomics-guided lifestyle intervention compared to a population-based (GLB/DPP) lifestyle intervention for weight management: results from the NOW randomised controlled trial. *BMJ Nutr Prev Health*. 2020;3:49–59. doi: 10.1136/bmjnp-2020-000073
32. Rein M, Ben-Yacov O, Godneva A, Shilo S, Zmora N, Kolobkov D, Cohen-Dolev N, Wolf BC, Kosower N, Lotan-Pompan M, et al. Effects of personalized diets by prediction of glycemic responses on glycemic control and metabolic health in newly diagnosed T2DM: a randomized dietary intervention pilot trial. *BMC Med*. 2022;20:56. doi: 10.1186/s12916-022-02254-y
33. Berry SE, Valdes AM, Drew DA, Asnicar F, Mazidi M, Wolf J, Capdevila J, Hadjiaggeorgiou G, Davies R, Al Khatib H, et al. Human postprandial responses to food and potential for precision nutrition. *Nat Med*. 2020;26:964–973. doi: 10.1038/s41591-020-0934-0
34. Birmingham KM, Linenberg I, Polidori L, Asnicar F, Arrè A, Wolf J, Badri F, Bernard H, Capdevila J, Buisiewicz WJ, et al. Effects of a personalized nutrition program on cardiometabolic health: a randomized controlled trial. *Nat Med*. 2024;30:1888–1897. doi: 10.1038/s41591-024-02951-6
35. Karvela M, Golden CT, Bell N, Martin-Li S, Bedzo-Nutakor J, Bosnic N, DeBeaudrap P, de Mateo-Lopez S, Alajrami A, Qin Y, et al. Assessment of the impact of a personalised nutrition intervention in impaired glucose regulation over 26 weeks: a randomised controlled trial. *Sci Rep*. 2024;14:5428. doi: 10.1038/s41598-024-55105-6
36. Ostchega Y, Berman LE, Hughes JP, Chen TC, Chiappa MM. Home blood pressure monitoring and hypertension status among US adults: the National Health and Nutrition Examination Survey (NHANES), 2009–2010. *Am J Hypertens*. 2013;26:1086–1092. doi: 10.1093/ajh/hpt054
37. Schutte AE, Kollias A, Stergiou GS. Blood pressure and its variability: classic and novel measurement techniques. *Nat Rev Cardiol*. 2022;19:643–654. doi: 10.1038/s41569-022-00690-0
38. Tan I, Gnanenthiran SR, Chan J, Kyriakoulis KG, Schlaich MP, Rodgers A, Stergiou GS, Schutte AE. Evaluation of the ability of a commercially available cuffless wearable device to track blood pressure changes. *J Hypertens*. 2023;41:1003–1010. doi: 10.1097/HJH.0000000000003428
39. Cross V, Stanford J, Gómez-Martín M, Collins CE, Robertson S, Clarke ED. Do personalized nutrition interventions improve dietary intake and risk factors in adults with elevated cardiovascular disease risk factors? A systematic review and meta-analysis of randomized controlled trials. *Nutr Rev*. 2025;83:e1709–e1721. doi: 10.1093/nutrit/nuae149
40. Appel LJ, Moore TJ, Obarzanek E, Vollmer WM, Svetkey LP, Sacks FM, Bray GA, Vogt TM, Cutler JA, Windhauser MM, et al. A clinical trial of the effects of dietary patterns on blood pressure. DASH Collaborative Research Group. *N Engl J Med*. 1997;336:1117–1124. doi: 10.1056/NEJM199704173361601
41. Appel LJ, Sacks FM, Carey VJ, Obarzanek E, Swain JF, Miller ER, Conlin PR, Erlinger TP, Rosner BA, Laranjo NM, et al; OmniHeart Collaborative Research Group. Effects of protein, monounsaturated fat, and carbohydrate intake on blood pressure and serum lipids: results of the OmniHeart randomized trial. *JAMA*. 2005;294:2455–2464. doi: 10.1001/jama.294.19.2455
42. Loo RL, Zou X, Appel LJ, Nicholson JK, Holmes E. Characterization of metabolic responses to healthy diets and association with blood pressure: application to the Optimal Macronutrient Intake Trial for Heart Health (Omni-Heart), a randomized controlled study. *Am J Clin Nutr*. 2018;107:323–334. doi: 10.1093/ajcn/nqx072
43. Aljuraiban GS, Griep LM, Chan Q, Daviglus ML, Stamler J, Van Horn L, Elliott P, Frost GS. Total, insoluble and soluble dietary fibre intake in relation to blood pressure: the INTERMAP Study. *Br J Nutr*. 2015;114:1480–1486. doi: 10.1017/S0007114515003098
44. Kawamura A, Kajiya K, Kishi H, Inagaki J, Mitarai M, Oda H, Umemoto S, Kobayashi S. Effects of the DASH-JUMP dietary intervention in Japanese participants with high-normal blood pressure and stage 1 hypertension: an open-label single-arm trial. *Hypertens Res*. 2016;39:777–785. doi: 10.1038/hr.2016.76
45. Baker EA, Barnidge EK, Schootman M, Sawicki M, Motton-Kershaw FL. Adaptation of a modified DASH diet to a rural African American community setting. *Am J Prev Med*. 2016;51:967–974. doi: 10.1016/j.amepre.2016.07.014
46. Nierenberg JL, Li C, He J, Gu D, Chen J, Lu X, Li J, Wu X, Gu CC, Hixson JE, et al. Blood pressure genetic risk score predicts blood pressure responses to dietary sodium and potassium: the GenSalt study (Genetic Epidemiology Network of Salt Sensitivity). *Hypertension*. 2017;70:1106–1112. doi: 10.1161/HYPERTENSIONAHA.117.10108
47. Juraschek SP, Miller ER III, Weaver CM, Appel LJ. Effects of sodium reduction and the DASH diet in relation to baseline blood pressure. *J Am Coll Cardiol*. 2017;70:2841–2848. doi: 10.1016/j.jacc.2017.10.011
48. Mill JG, Baldo MP, Molina MCB, Schmidt MI, Barreto SM, Chor D, Griep RH, Matos SM, Ribeiro ALP, Duncan BB, et al. Sex-specific patterns in the association between salt intake and blood pressure: the ELISA-Brasil study. *J Clin Hypertens (Greenwich)*. 2019;21:502–509. doi: 10.1111/jch.13509
49. Huat J, Cirillo A, Taminiau B, Descy J, Saint-Remy A, Daube G, Krzesinski JM, Melin P, de Tullio P, Jouret F. Human stool metabolome differs upon 24 h blood pressure levels and blood pressure dipping status: a prospective longitudinal study. *Metabolites*. 2021;11:282. doi: 10.3390/metabo11050282
50. Wang Y, Feng L, Zeng G, Zhu H, Sun J, Gao P, Yuan J, Lan X, Li S, Zhao Y, et al; DECIDE-Diet Study Group. Effects of cuisine-based Chinese heart-healthy diet in lowering blood pressure among adults in China: multicenter, single-blind, randomized, parallel controlled feeding trial. *Circulation*. 2022;146:303–315. doi: 10.1161/CIRCULATIONAHA.122.059045
51. Reay WR, Clarke E, Eslick S, Riveros C, Holliday EG, McEvoy MA, Peel R, Hancock S, Scott RJ, Attia JR, et al. Using genetics to inform interventions related to sodium and potassium in hypertension. *Circulation*. 2024;149:1019–1032. doi: 10.1161/CIRCULATIONAHA.123.065394
52. Kallapura G, Prakash AS, Sankaran K, Manjappa P, Chaudhary P, Ambhore S, Dhar D. Microbiota based personalized nutrition improves hyperglycaemia and hypertension parameters and reduces inflammation: a prospective,

- open label, controlled, randomized, comparative, proof of concept study. *PeerJ*. 2024;12:e17583. doi: 10.7717/peerj.17583
53. Verhaar BJH, Wijdeveld M, Wortelboer K, Rampanelli E, Levels JHM, Collard D, Cammenga M, Nageswaran V, Haghighi A, Landmesser U, et al. Effects of oral butyrate on blood pressure in patients with hypertension: a randomized, placebo-controlled trial. *Hypertension*. 2024;81:2124–2136. doi: 10.1161/HYPERTENSIONAHA.123.22437
 54. Sun Y, Zhang R, Tian L, Pan Y, Sun X, Huang Z, Fan J, Chen J, Zhang K, Li S, et al. Novel metabolites associated with blood pressure after dietary interventions. *Hypertension*. 2024;81:1966–1975. doi: 10.1161/HYPERTENSIONAHA.124.22999
 55. Unger T, Borghi C, Charchar F, Khan NA, Poulter NR, Prabhakaran D, Ramirez A, Schlaich M, Stergiou GS, Tomaszewski M, et al. 2020 International Society of Hypertension global hypertension practice guidelines. *Hypertension*. 2020;75:1334–1357. doi: 10.1161/HYPERTENSIONAHA.120.15026
 56. Cook NR, Appel LJ, Whelton PK. Lower levels of sodium intake and reduced cardiovascular risk. *Circulation*. 2014;129:981–989. doi: 10.1161/CIRCULATIONAHA.113.006032
 57. Whelton PK, Appel L, Charleston J, Dalcin AT, Ewart C, Fried L, Kaidy D, Klag MJ, Kumanyika S, Steffen L, et al. The effects of nonpharmacologic interventions on blood pressure of persons with high normal levels: results of the trials of hypertension prevention, phase I. *JAMA*. 1992;267:1213–1220. doi: 10.1001/jama.1992.03480090061028
 58. Appel LJ, Espeland MA, Easter L, Wilson AC, Folmar S, Lacy CR. Effects of reduced sodium intake on hypertension control in older individuals: results from the Trial of Nonpharmacologic Interventions in the Elderly (TONE). *Arch Intern Med*. 2001;161:685–693. doi: 10.1001/archinte.161.5.685
 59. Chan Q, Stamler J, Griep LM, Daviglus ML, Horn LV, Elliott P. An update on nutrients and blood pressure. *J Atheroscler Thromb*. 2016;23:276–289. doi: 10.5551/jat.30000
 60. 2015–2020 Dietary Guidelines for Americans. U.S. Department of Health and Human Services and U.S. Department of Agriculture; 2015.
 61. Svetkey LP, Sacks FM, Obarzanek E, Vollmer WM, Appel LJ, Lin PH, Karanja NM, Harsha DW, Bray GA, Aickin M, et al. The DASH Diet, Sodium Intake and Blood Pressure Trial (DASH-sodium): rationale and design. DASH-Sodium Collaborative Research Group. *J Am Diet Assoc*. 1999;99:S96–104. doi: 10.1016/s0002-8223(99)00423-x
 62. LeBlanc-Morales N. Culinary medicine: patient education for therapeutic lifestyle changes. *Crit Care Nurs Clin North Am*. 2019;31:109–123. doi: 10.1016/j.cnc.2018.11.009
 63. Ozemek C, Phillips SA, Popovic D, Laddu-Patel D, Fancher IS, Arena R, Lavie CJ. Nonpharmacologic management of hypertension: a multidisciplinary approach. *Curr Opin Cardiol*. 2017;32:381–388. doi: 10.1097/HCO.0000000000000406
 64. Aljuraiban GS, Gibson R, Chan DS, Van Horn L, Chan Q. The role of diet in the prevention of hypertension and management of blood pressure: an umbrella review of meta-analyses of interventional and observational studies. *Adv Nutr*. 2024;15:100123. doi: 10.1016/j.advnut.2023.09.011
 65. Monsivais P, Rehm CD, Drewnowski A. The DASH diet and diet costs among ethnic and racial groups in the United States. *JAMA Intern Med*. 2013;173:1922–1924. doi: 10.1001/jamainternmed.2013.9479
 66. Jama HA, Snelson M, Schutte AE, Muir J, Marques FZ. Recommendations for the use of dietary fiber to improve blood pressure control. *Hypertension*. 2024;81:1450–1459. doi: 10.1161/HYPERTENSIONAHA.123.22575
 67. Filippini T, Naska A, Kasdagli MI, Torres D, Lopes C, Carvalho C, Moreira P, Malavolti M, Orsini N, Whelton PK, et al. Potassium intake and blood pressure: a dose-response meta-analysis of randomized controlled trials. *J Am Heart Assoc*. 2020;9:e015719. doi: 10.1161/JAHA.119.015719
 68. Neal B, Wu Y, Feng X, Zhang R, Zhang Y, Shi J, Zhang J, Tian M, Huang L, Li Z, et al. Effect of salt substitution on cardiovascular events and death. *N Engl J Med*. 2021;385:1067–1077. doi: 10.1056/NEJMoa2105675
 69. Chan RJ, Parikh N, Ahmed S, Ruzicka M, Hiremath S. Blood pressure control should focus on more potassium: controversies in hypertension. *Hypertension*. 2024;81:501–509. doi: 10.1161/HYPERTENSIONAHA.123.20545
 70. Jones DW, Ferdinand KC, Taler SJ, Johnson HM, Shimbo D, Abdalla M, Altieri MM, Bansal N, Bello NA, Bress AP, et al; Writing Committee Members. 2025 AHA/ACC/AANP/AAPA/ABC/ACCP/ACPM/AGS/AMA/ASPC/NMA/PCNA/SGIM Guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2025;152:e114–e218. doi: 10.1161/CIR.0000000000001356
 71. Yin X, Rodgers A, Perkovic A, Huang L, Li KC, Yu J, Wu Y, Wu JHY, Marklund M, Huffman MD, et al. Effects of salt substitutes on clinical outcomes: a systematic review and meta-analysis. *Heart*. 2022;108:1608–1615. doi: 10.1136/heartjnl-2022-321332
 72. Ingelfinger JR. Monogenic and polygenic contributions to hypertension. In: Flynn JT, Ingelfinger JR, Brady T, eds. *Pediatric Hypertension*. Springer International Publishing; 2020:1–20.
 73. Luft FC. Personal genetic-hypertension odyssey from phenotypes to genotypes and targets. *Hypertension*. 2024;81:2395–2406. doi: 10.1161/HYPERTENSIONAHA.124.21714
 74. Keaton JM, Kamali Z, Xie T, Vaez A, Williams A, Goleva SB, Ani A, Evangelou E, Hellwege JN, Yengo L, et al; Million Veteran Program. Genome-wide analysis in over 1 million individuals of European ancestry yields improved polygenic risk scores for blood pressure traits. *Nat Genet*. 2024;56:778–791. doi: 10.1038/s41588-024-01714-w
 75. Padmanabhan S, Dominiczak AF. Genomics of hypertension: the road to precision medicine. *Nat Rev Cardiol*. 2021;18:235–250. doi: 10.1038/s41569-020-00466-4
 76. Liu R, Mi B, Zhao Y, Li Q, Dang S, Yan H. Gender-specific association between carbohydrate consumption and blood pressure in Chinese adults. *BMJ Nutr Prev Health*. 2021;4:80–89. doi: 10.1136/bmjnp-2020-000165
 77. Masenga SK, Kirabo A, Hamooya BM, Nzala S, Kwenda G, Heimbürger DC, Mutale W, Koethe JR, Pilic L, Munsaka SM. HIV-positive demonstrate more salt sensitivity and nocturnal non-dipping blood pressure than HIV-negative individuals. *Clin Hypertens*. 2021;27:2. doi: 10.1186/s40885-020-00160-0
 78. Masenga SK, Hamooya BM, Nzala S, Kwenda G, Heimbürger DC, Mutale W, Munsaka SM, Koethe JR, Kirabo A. Patho-immune mechanisms of hypertension in HIV: a systematic and thematic review. *Curr Hypertens Rep*. 2019;21:56. doi: 10.1007/s11906-019-0956-5
 79. Leng B, Jin Y, Li G, Chen L, Jin N. Socioeconomic status and hypertension: a meta-analysis. *J Hypertens*. 2015;33:221–229. doi: 10.1097/HJH.0000000000000428
 80. Ciarambino T, Crispino P, Guarisco G, Giordano M. Gender differences in insulin resistance: new knowledge and perspectives. *Curr Issues Mol Biol*. 2023;45:7845–7861. doi: 10.3390/cimb45100496
 81. Lee CMY, Huxley RR, Wildman RP, Woodward M. Indices of abdominal obesity are better discriminators of cardiovascular risk factors than BMI: a meta-analysis. *J Clin Epidemiol*. 2008;61:646–653. doi: 10.1016/j.jclinepi.2007.08.012
 82. Bennett E, Peters SAE, Woodward M. Sex differences in macronutrient intake and adherence to dietary recommendations: findings from the UK Biobank. *BMJ Open*. 2018;8:e020017. doi: 10.1136/bmjopen-2017-020017
 83. Carey RM, Schoeffel CD, Gildea JJ, Jones JE, McGrath HE, Gordon LN, Park MJ, Sobota RS, Underwood PC, Williams J, et al. Salt sensitivity of blood pressure is associated with polymorphisms in the sodium-bicarbonate cotransporter. *Hypertension*. 2012;60:1359–1366. doi: 10.1161/HYPERTENSIONAHA.112.196071
 84. Wang Y, Zhou Q, Gao WH, Yan Y, Chu C, Chen C, Yuan Y, Wang KK, Ma Q, Gao K, et al. Association of plasma cyclooxygenase-2 levels and genetic polymorphisms with salt sensitivity, blood pressure changes and hypertension incidence in Chinese adults. *J Hypertens*. 2020;38:1745–1754. doi: 10.1097/HJH.0000000000002473
 85. Kurtz TW, DiCarlo SE, Pravenec M, Morris RC. An appraisal of methods recently recommended for testing salt sensitivity of blood pressure. *J Am Heart Assoc*. 2017;6:e005653. doi: 10.1161/JAHA.117.005653
 86. O'Donnell JA, Zheng T, Meric G, Marques FZ. The gut microbiome and hypertension. *Nat Rev Nephrol*. 2023;19:153–167. doi: 10.1038/s41581-022-00654-0
 87. Snelson M, Liu CF, Markó L, Forslund SK, Marques FZ, Tang WHW. Gut-heart axis: the role of gut microbiota and metabolites in heart failure. *Circ Res*. 2025;136:1382–1406. doi: 10.1161/circresaha.125.325516
 88. Korpela K, Flint HJ, Johnstone AM, Lappi J, Poutanen K, Dewulf EM, Delzenne NM, de Vos WM, Salonen A. Gut microbiota signatures predict host and microbiota responses to dietary interventions in obese individuals. *PLoS One*. 2014;9:e90702. doi: 10.1371/journal.pone.0090702
 89. Rothschild D, Weissbrod O, Barkan E, Kurilshikov A, Korem T, Zeevi D, Costea PI, Godneva A, Kalka IN, Bar N, et al. Environment dominates over host genetics in shaping human gut microbiota. *Nature*. 2018;555:210–215. doi: 10.1038/nature25973
 90. Opperman C, Majzoobi M, Farahnaky A, Shah R, Van TTH, Ratanpaul V, Blanch EW, Brennan C, Eri R. Beyond soluble and insoluble: a comprehensive framework for classifying dietary fibre's health effects. *Food Res Int*. 2025;206:115843. doi: 10.1016/j.foodres.2025.115843
 91. Khan K, Jovanovski E, Ho HVT, Marques ACR, Zurbau A, Mejia SB, Sievenpiper JL, Vuksan V. The effect of viscous soluble fiber on blood pressure: a systematic review and meta-analysis of randomized controlled trials. *Nutr Metab Cardiovasc Dis*. 2018;28:3–13. doi: 10.1016/j.numecd.2017.09.007

92. Marques FZ, Nelson E, Chu PY, Horlock D, Fiedler A, Ziemann M, Tan JK, Kuruppu S, Rajapakse NW, El-Osta A, et al. High-fiber diet and acetate supplementation change the gut microbiota and prevent the development of hypertension and heart failure in hypertensive mice. *Circulation*. 2017;135:964–977. doi: 10.1161/CIRCULATIONAHA.116.024545
93. Kaye DM, Shihata W, Jama HA, Tsyganov K, Ziemann M, Kiriazis H, Horlock D, Vijay A, Giam B, Vinh A, et al. Deficiency of prebiotic fibre and insufficient signalling through gut metabolite sensing receptors leads to cardiovascular disease. *Circulation*. 2020;141:1393–1403. doi: 10.1161/CIRCULATIONAHA.119.043081
94. Kim S, Goel R, Kumar A, Qi Y, Lobaton G, Hosaka K, Mohammed M, Handberg EM, Richards EM, Pepine CJ, et al. Imbalance of gut microbiome and intestinal epithelial barrier dysfunction in patients with high blood pressure. *Clin Sci (Lond)*. 2018;132:701–718. doi: 10.1042/CS20180087
95. Jama HA, Rhys-Jones D, Nakai M, Yao CK, Climie RE, Sata Y, Anderson D, Creek DJ, Head GA, Kaye DM, et al. Prebiotic intervention with HAMSAB in untreated essential hypertensive patients assessed in a phase II randomized trial. *Nat Cardiovasc Res*. 2023;2:35–43. doi: 10.1038/s44161-022-00197-4
96. Bartolomeus H, Balogh A, Yakoub M, Homann S, Markó L, Höges S, Tsvetkov D, Krannich A, Wundersitz S, Avery EG, et al. Short-chain fatty acid propionate protects from hypertensive cardiovascular damage. *Circulation*. 2019;139:1407–1421. doi: 10.1161/CIRCULATIONAHA.118.036652
97. Dinakis E, Xu C, Muralitharan RR, Jama H, Xie L, Leung KC, Mirabito Colafella KM, McArdle Z, Salimova E, Camargo Tavares L, et al. pH-sensor GPR68 plays a role in how dietary fibre lowers blood pressure in a preclinical model of hypertension. *Clin Sci (Lond)*. 2025;139:1239–1259. doi: 10.1042/cs20243009
98. Wu G, Xu T, Zhao N, Lam YY, Ding X, Wei D, Fan J, Shi Y, Li X, Li M, et al. A core microbiome signature as an indicator of health. *Cell*. 2024;187:6550–6565.e11. doi: 10.1016/j.cell.2024.09.019
99. Calderón-Pérez L, Gosalbes MJ, Yuste S, Valls RM, Pedret A, Llauro E, Jimenez-Hernandez N, Artacho A, Pla-Pagà L, Companys J, et al. Gut metagenomic and short chain fatty acids signature in hypertension: a cross-sectional study. *Sci Rep*. 2020;10:6436. doi: 10.1038/s41598-020-63475-w
100. Yan Q, Gu Y, Li X, Yang W, Jia L, Chen C, Han X, Huang Y, Zhao L, Li P, et al. Alterations of the gut microbiome in hypertension. *Front Cell Infect Microbiol*. 2017;7:381. doi: 10.3389/fcimb.2017.00381
101. Verhaar BJH, Collard D, Prodan A, Levels JHM, Zwinderman AH, Bäckhed F, Vogt L, Peters MJL, Muller M, Nieuwdorp M, et al. Associations between gut microbiota, faecal short-chain fatty acids, and blood pressure across ethnic groups: the HELIUS study. *Eur Heart J*. 2020;41:4259–4267. doi: 10.1093/eurheartj/ehaa704
102. Snelson M, Rhys-Jones D, Jama HA, Creek DJ, Mackay CR, Muir J, Marques FZ. Preintervention intake of whole grains versus refined grains, and the gut microbiome, discriminate the antihypertensive effect of prebiotic fiber. *Circ Genom Precis Med*. 2025;18:e005019. doi: 10.1161/CIRCGEN.124.005019
103. Gill PA, Muir JG, Gibson PR, van Zelm MC. A randomized dietary intervention to increase colonic and peripheral blood SCFAs modulates the blood B- and T-cell compartments in healthy humans. *Am J Clin Nutr*. 2022;116:1354–1367. doi: 10.1093/ajcn/nqac246
104. Gut microbial metabolites lower blood pressure in patients with hypertension. *Nat Cardiovasc Res*. 2023;2:18–19. doi: 10.1038/s44161-022-00204-8
105. Camargo Tavares L, Marques FZ. Clinical trial evidence of the gut microbial metabolite butyrate in hypertension. *Hypertension*. 2024;81:2137–2139. doi: 10.1161/HYPERTENSIONAHA.124.23736
106. Pituch A, Walkowiak J, Banaszkiewicz A. Butyric acid in functional constipation. *Prz Gastroenterol*. 2013;8:295–298. doi: 10.5114/pg.2013.38731
107. Vashisht E, Nguyen H, Marques FZ, Snelson M. Diet, gut microbiota, and intestinal permeability: emerging mechanisms in hypertension pathogenesis. *Can J Cardiol*. 2026;42:1818–1831. doi: 10.1016/j.cjca.2026.03.005
108. Spencer SP, Fragiadakis GK, Sonnenburg JL. Pursuing human-relevant gut microbiota-immune interactions. *Immunity*. 2019;51:225–239. doi: 10.1016/j.immuni.2019.08.002
109. Nakai M, Ribeiro RV, Stevens BR, Gill P, Muralitharan RR, Yiallourou S, Muir J, Carrington M, Head GA, Kaye DM, et al. Essential hypertension is associated with changes in gut microbial metabolic pathways: a multisite analysis of ambulatory blood pressure. *Hypertension*. 2021;78:804–815. doi: 10.1161/HYPERTENSIONAHA.121.17288
110. Muralitharan R R, Zheng T, Dinakis E, Xie L, Barbaro-Wahl A, Jama HA, Nakai M, Paterson M, Leung KC, McArdle Z, et al. Gut microbiota metabolites sensed by host GPR41/43 protect against hypertension. *Circ Res*. 2025;136:e20–e33. doi: 10.1161/CIRCRESAHA.124.325770
111. Tavares LC, Muralitharan RR, Snelson M, Marques FZ. Rare pathogenic variants in G-protein-coupled receptor genes involved in gut-to-host communication are associated with cardiovascular disease risk. *Cardiovasc Res*. 2025;121:1316. doi: 10.1093/cvr/cvaf070
112. Al Ashmar S, Anwardeen NR, Anlar GG, Pedersen S, Elrayess MA, Zeidan A. Metabolomic profiling reveals key metabolites associated with hypertension progression. *Front Cardiovasc Med*. 2024;11:1284114. doi: 10.3389/fcvm.2024.1284114
113. Lin YT, Salihovic S, Fall T, Hammar U, Ingelsson E, Årnlöv J, Lind L, Sundström J. Global plasma metabolomics to identify potential biomarkers of blood pressure progression. *Arterioscler Thromb Vasc Biol*. 2020;40:e227–e237. doi: 10.1161/ATVBAHA.120.314356
114. Shi M, He J, Li C, Lu X, He WJ, Cao J, Chen J, Chen JC, Bazzano LA, Li JX, et al. Metabolomics study of blood pressure salt-sensitivity and hypertension. *Nutr Metab Cardiovasc Dis*. 2022;32:1681–1692. doi: 10.1016/j.numecd.2022.04.002
115. Rebholz CM, Lichtenstein AH, Zheng Z, Appel LJ, Coresh J. Serum untargeted metabolomic profile of the Dietary Approaches to Stop Hypertension (DASH) dietary pattern. *Am J Clin Nutr*. 2018;108:243–255. doi: 10.1093/ajcn/nqy099
116. Han JM, Guo L, Chen XH, Xie Q, Song XY, Ma YL. Relationship between trimethylamine N-oxide and the risk of hypertension in patients with cardiovascular disease: a meta-analysis and dose-response relationship analysis. *Medicine (Baltimore)*. 2024;103:e36784. doi: 10.1097/MD.00000000000036784
117. Dinakis E, O'Donnell JA, Marques FZ. The gut-immune axis during hypertension and cardiovascular diseases. *Acta Physiol*. 2024;240:e14193. doi: 10.1111/apha.14193
118. Roberts AB, Gu X, Buffa JA, Hurd AG, Wang Z, Zhu W, Gupta N, Skye SM, Cody DB, Levison BS, et al. Development of a gut microbe-targeted non-lethal therapeutic to inhibit thrombosis potential. *Nat Med*. 2018;24:1407–1417. doi: 10.1038/s41591-018-0128-1
119. Wang Z, Roberts AB, Buffa JA, Levison BS, Zhu W, Org E, Gu X, Huang Y, Zamanian-Daryoush M, Culley MK, et al. Non-lethal inhibition of gut microbial trimethylamine production for the treatment of atherosclerosis. *Cell*. 2015;163:1585–1595. doi: 10.1016/j.cell.2015.11.055
120. Hsu CN, Hou CY, Lee CT, Chang-Chien GP, Lin S, Tain YL. Maternal 3,3-dimethyl-1-butanol therapy protects adult male rat offspring against hypertension programmed by perinatal TCDD exposure. *Nutrients*. 2021;13:3041. doi: 10.3390/nu13093041
121. Abdill RJ, Adamowicz EM, Blekman R. Public human microbiome data are dominated by highly developed countries. *PLoS Biol*. 2022;20:e3001536. doi: 10.1371/journal.pbio.3001536
122. Wuni R, Vimalaswaran KS. Barriers in translating existing nutrigenetics insights to precision nutrition for cardiometabolic health in ethnically diverse populations. *Lifestyle Genom*. 2024;17:122–135. doi: 10.1159/000541909
123. Collins J, Bertrand B, Hayes V, Li SX, Thomas J, Truby H, Whelan K. The application of genetics and nutritional genomics in practice: an international survey of knowledge, involvement and confidence among dietitians in the US, Australia and the UK. *Genes Nutr*. 2013;8:523–533. doi: 10.1007/s12263-013-0351-9
124. Roosan D, Wu Y, Tran M, Huang Y, Baskys A, Roosan MR. Opportunities to integrate nutrigenomics into clinical practice and patient counseling. *Eur J Clin Nutr*. 2023;77:36–44. doi: 10.1038/s41430-022-01146-x
125. Carrillo-Alvarez E, Rifà-Ros R, Salinas-Roca B, Costa-Tutusaus L, Lamas M, Rodríguez-Monforte M. Diet-related health inequalities in high-income countries: a scoping review of observational studies. *Adv Nutr*. 2025;16:100439. doi: 10.1016/j.advnut.2025.100439
126. Khoury MJ, Bowen S, Dotson WD, Drzymalla E, Green RF, Goldstein R, Kolor K, Liburd LC, Sperling LS, Bunnell R. Health equity in the implementation of genomics and precision medicine: a public health imperative. *Genet Med*. 2022;24:1630–1639. doi: 10.1016/j.jim.2022.04.009
127. Spruill TM, Muntner P, Popp CJ, Shimbo D, Cooper LA, Moran AE, Penko J, Bibbins-Domingo K, Ibe C, Nnodim Opara I, et al. Addressing social determinants to prevent hypertension (The RESTORE Network): overview of the health equity research network to prevent hypertension. *Am J Hypertens*. 2023;36:232–239. doi: 10.1093/ajh/hpad010
128. Karnik A, Yang E. Artificial intelligence to improve blood pressure control: a state-of-the-art review. *Am J Hypertens*. 2025;38:519–526. doi: 10.1093/ajh/hpaf035
129. Dombrowski SU, Avenell A, Sniehot FF. Behavioural interventions for obese adults with additional risk factors for morbidity: systematic review of effects on behaviour, weight and disease risk factors. *Obes Facts*. 2010;3:377–396. doi: 10.1159/000323076
130. Institute of Medicine. Implementing a population-based policy and systems approach to the prevention and control of hypertension. In: *A Population-Based Policy and Systems Change Approach to Prevent and Control Hypertension*. The National Academies Press; 2010